Evaluating Operational Efficiency at Hartsfield-Jackson Atlanta International Airport (KATL)

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I. Abstract

This study analyzes the efficiency of Hartsfield-Jackson Atlanta International Airport (KATL) using a 2024 sample of 496 flights originating at KATL. A One-Way Analysis of Variance found that there was no statistically significant difference in mean departure delays across weekdays. A Multiple Linear Regression model showed that departure delay is a significant predictor of arrival delay and that route distance is not a significant predictor of arrival delay. Lastly, a non-parametric Kruskal-Wallis test confirmed that departure delay distributions shift significantly across operating carriers at KATL. It is indicated that operational efficiency and resiliency are determined by carrier-specific logistics rather than airport-wide infrastructure or temporal factors.

II. Introduction

The dataset analyzed in this study was sourced from Kaggle, and it was originally compiled by the U.S. Bureau of Transportation Statistics (BTS). The dataset provides flight operation information for domestic flights originating at KATL in 2024. The dataset includes information on delay times, cancellations, route distances, diversions, etc. Monthly csv files for January-December 2024 were downloaded from the BTS TranStats On-Time Performance database, and 35 relevant columns were selected. The monthly files were merged into a single dataset. The original dataset contained over 7 million observations. A sample of 10,000 was downloaded, and after filtering the dataset to only include flights that originated at KATL, a sample of 496 was used for this analysis. I found this dataset interesting because KATL is the busiest airport in the world, and I wanted to get insights on its efficiency.

Table 1. Variable Descriptions

Variable Name	Description	Variable Type	Units/Label
day_name	The day of the week on which a flight departed (relabeling of day_of_week variable)	Categorical	1 = Monday 7 = Sunday
op_unique_carrier	Unique alpha code identifying the operating airline	Categorical	DL = Delta Air Lines, AA = American Airlines, etc.
dep_delay	Length of departure delay	Quantitative	Minutes negative = early departure 0 = on time departure positive = late departure
arr_delay	Length of arrival delay	Quantitative	Minutes negative = early arrival 0 = on time arrival positive = late arrival
distance	Distance between origin city and destination city	Quantitative	Miles

III. Data Overview

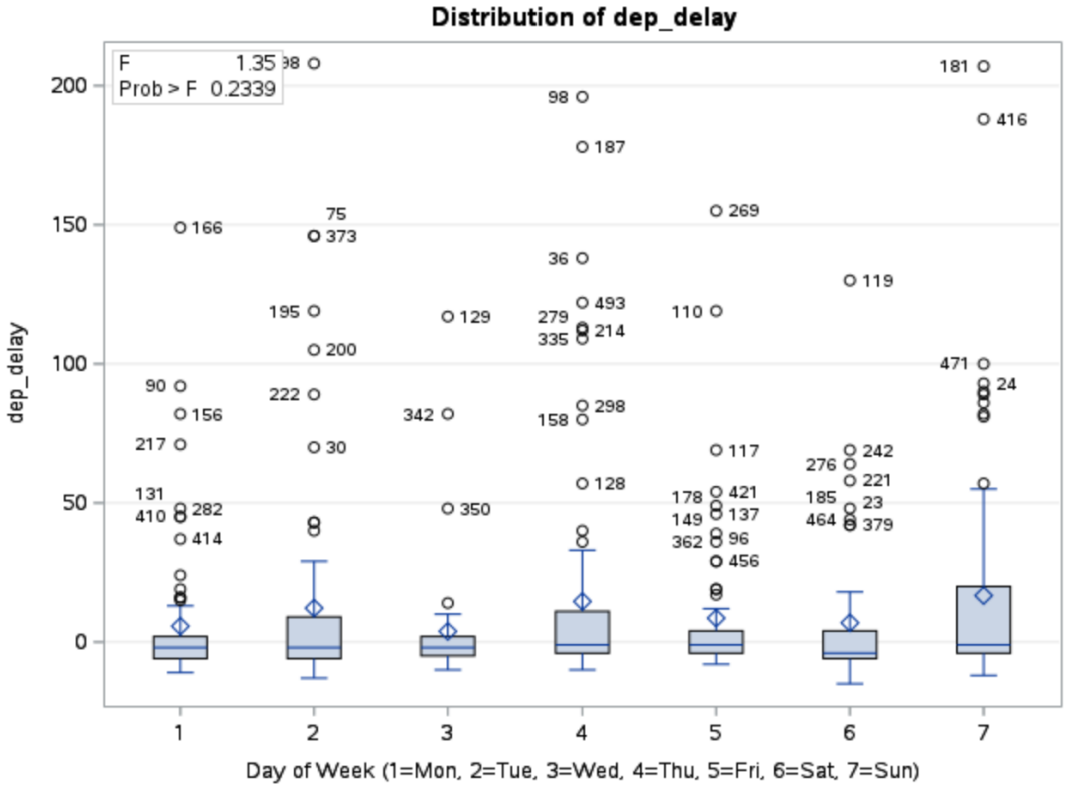


Figure 1. Distribution of Departure Delays Across Days of the Week – The median departure delay (horizontal lines inside the boxes) for all seven days of the week falls below zero minutes indicating early departures. The mean departure delay (diamond symbols) for all seven days falls above the zero-minute mark indicating late departures. Severe positive outliers are present across all seven days.

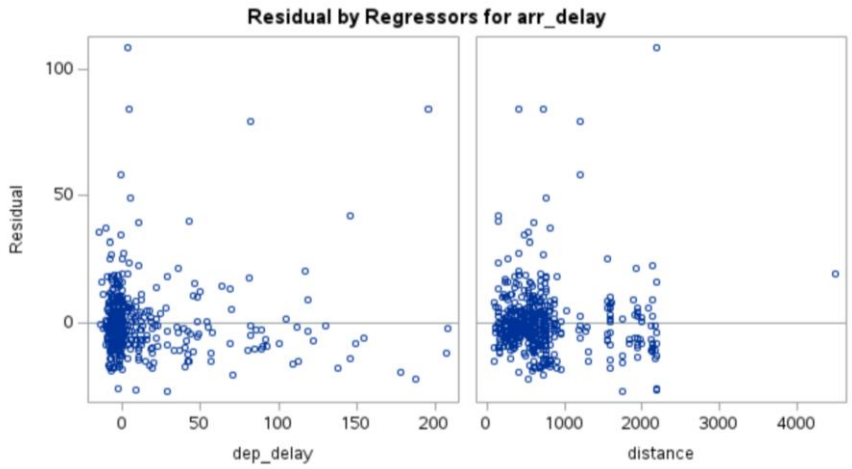


Figure 2. Residual vs. Departure Delay and Route Distance – The chart on the left plots the error residuals against departure delay. Data points are densely clustered around the zero-minute mark. This reflects a large proportion of on-time flights. The vertical spread of data points changes as departure delay increases. The chart on the right plots the error residuals against route distance. The data points are densely packed between zero and 1,000 miles, reflecting the high proportion of domestic flights departing from KATL.

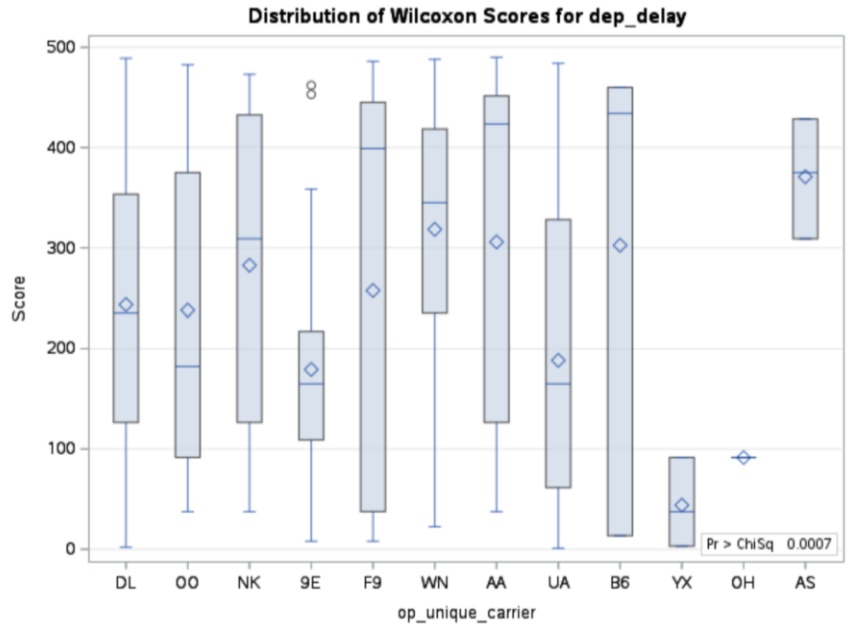


Figure 3. Distribution of Wilcoxon Scores for Departure Delay Across Operating Carriers – The chart shows the varying distributions of departure delays across twelve operating carriers at KATL. Most carriers, including Delta Air Lines (DL) and Frontier Airlines (F9), had a lot of variability in departure delay times. Carriers such as Republic Airways (YX) and Alaska Airlines (AS) did not have a high amount of variability in departure delay times. Overall, delay patterns differ greatly between the airlines.

IV. Analysis and Results (an alpha level of α = 0.05 was used for all tests)

A. Hypothesis 1: Flight departure delays are significantly higher on certain days of the week (One-Way ANOVA)

H_0: There is no difference in the population mean departure delays of each day of the week

H_A: At least one day of the week has a population mean departure delay that is different from the others

Weekday flight efficiency was analyzed using a One-Way ANOVA (Table 2). The categorical variable is day_name (day of the week, 1 = Monday, 7 = Sunday), and the quantitative variable is dep_delay (departure delay measured in minutes). The procedure tests whether the average departure delay is different for each of the seven days of the week. Because the calculated p-value (0.2339) is greater than the significance level of 0.05, I fail to reject the null hypothesis. There is not enough evidence to suggest a statistically significant difference in the population mean length of departure delay for the different days of the week. KATL maintains a stable baseline throughout the week.

Table 2. One-Way ANOVA Results – There is not a statistically significant difference in the population mean departure delay between the days of the week. The average departure delay was 10.5 minutes for all days of the week. Thursday had the highest average departure delay, and Wednesday had the lowest average departure delay.

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	9436.3237	1572.7206	1.35	0.2339
Error	483	563265.2763	1166.1807		
Corrected Total	489	572701.6000			

R-Square	Coeff Var	Root MSE	dep_delay Mean
0.016477	326.5652	34.14939	10.45714

Source	DF	Anova SS	Mean Square	F Value	Pr > F
day_of_week	6	9436.323671	1572.720612	1.35	0.2339

Level of day_of_week	N	dep_delay	
		Mean	Std Dev
1	77	5.6883117	26.0451328
2	78	12.1923077	39.8174594
3	45	3.8222222	22.9805144
4	90	14.5888889	40.7182411
5	69	8.5797101	27.6705438
6	54	6.8888889	26.3278794
7	77	16.7012987	41.3639383

B. Hypothesis 2: Departure delay and route distance are predictors of arrival delay (Multiple Linear Regression)

H_0: Both departure delay and route distance slope coefficients are equal to zero

H_A: At least one slope coefficient is not equal to zero

To evaluate whether departure delay and route distance are predictors of arrival delay, a Multiple Linear Regression model was developed (Table 3). The Response variable was arr_delay (arrival delay in minutes), and the independent variables were dep_delay (departure delay in minutes) and distance (distance between origin city and destination city). Because the model had a p-value less than 0.0001, I rejected the null hypothesis. There is enough evidence to suggest that at least one slope coefficient is not equal to zero. Individually, departure delay had a p-value less than 0.0001 and a positive slope coefficient of 1.05. This means that for every 1-minute increase in departure delay, arrival delay increases by about 1.05 minutes. With a p-value greater than the significance level of 0.05, departure delay is a significant predictor of arrival delay. Route distance had a p-value of 0.6450, indicating that it is not a statistically significant predictor of arrival delay.

Table 3. Multiple Linear Regression Results – The model yielded a p-value less than 0.0001 and an R^2 value of 0.8807. 88% of the variation in arrival delay is explained by the model.

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	2	628677	314338	1786.04	<.0001
Error	484	85182	175.99689		
Corrected Total	486	713859			

Root MSE	13.26638	R-Square	0.8807
Dependent Mean	6.46201	Adj R-Sq	0.8802
Coeff Var	205.29800		

Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	-4.87234	1.00940	-4.83	<.0001
dep_delay	dep_delay	1	1.04756	0.01754	59.74	<.0001
distance	distance	1	0.00051770	0.00112	0.46	0.6450

C. Hypothesis 3: Different operating carriers have different departure delay distributions (Kruskal-Wallis Test)

H_0: All operating carriers have the same population median departure delay

H_A: At least one operating carrier has a population median departure delay that is different from the others

A non-parametric Kruskal-Wallis test was conducted to examine the distribution of departure delays between twelve different operating carriers at KATL (Table 4). The independent variable was op_unique_carrier (unique airline code), and the dependent variable was dep_delay (departure delay in minutes). The p-value for the test was 0.0007. Because the p-value was less than the alpha level of 0.05, I rejected the null hypothesis. There is enough evidence to suggest that at least one carrier has a population median departure delay that is different from the others. Most carriers exhibit a lot of variability in length of departure delays.

Table 4. Kruskal-Wallis Test Results – Highly significant result (p-value = 0.0007) indicates that delay patterns differ among the airlines.

Wilcoxon Scores (Rank Sums) for Variable dep_delay Classified by Variable op_unique_carrier					
op_unique_carrier	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
DL	327	79583.50	80278.50	1474.51758	243.374618
OO	22	5234.00	5401.00	648.06198	237.909091
NK	17	4804.00	4173.50	572.71335	282.588235
9E	32	5737.50	7856.00	773.19672	179.296875
F9	23	5920.00	5646.50	661.91867	257.391304
WN	39	12417.50	9574.50	847.03791	318.397436
AA	5	1529.00	1227.50	314.51267	305.800000
UA	15	2826.00	3682.50	539.10668	188.400000
B6	3	907.50	736.50	244.12226	302.500000
YX	3	132.00	736.50	244.12226	44.000000
OH	1	91.50	245.50	141.23317	91.500000
AS	3	1112.50	736.50	244.12226	370.833333
Average scores were used for ties.					

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
32.3381	11	0.0007

V. Conclusion

This study evaluated the operational efficiency of flights departing from Hartsfield-Jackson Atlanta International Airport in 2024. Together, the results of the One-Way ANOVA and Kruskal-Wallis tests revealed a crucial insight. KATL as a whole does not experience more delays on certain days of the week. Different operating carriers experience more delays than others. While the general infrastructure of the airport is able to handle traffic smoothly throughout the week regardless of increases in demand, there is variability in the operations of the different carriers. The Multiple Linear Regression model revealed that the length of an initial departure delay is almost directly proportional to the length of the arrival delay. A flight departing from KATL 10 minutes late can be expected to arrive at its destination about 10 minutes late as well. The regression model also revealed that the distance between an origin city and a destination city is not a significant predictor of arrival delay. Longer flights don't have longer delays and shorter flights don't have shorter delays. Overall, these findings indicate that operational efficiency at KATL may be due to differences in carrier-specific logistics rather than airport-wide infrastructure, timing, or geographic location in relation to destination airports.